

TECHNICAL ASSESSMENT REPORT

Feasibility Analysis: Magnetometer-Based Detection Systems for Vietnam-Era Aircraft Wreckage Recovery Operations

Date: October 24, 2025
To: Nathan Strachen, Assistant Teaching Professor
CC: Ahmad (Research Team), Ed (Vietnam Pilots Liaison),
Re: Magnetometer Detection of Ferromagnetic Materials in Vietnam-Era Aircraft

EXECUTIVE BRIEF

Finding: Magnetometer-based detection systems offer a viable, cost-effective solution for locating Vietnam-era aircraft wreckage in coastal waters. Field data confirms sufficient ferromagnetic signatures for reliable detection at operationally useful ranges.

Key Technical Parameters:

- Ferromagnetic content: 2,500-4,000 kg per fighter aircraft (primarily engines and landing gear)
- Detection range: 40-70 meters for complete aircraft, 25-45 meters for major components
- System cost: \$20K-60K for operational capability (vastly cheaper than diving operations)
- Coverage rate: 2-3 km²/day at 25m line spacing with 90%+ detection probability
- Proven success: DPAA/Project Recover B-52 recovery (2020) validates methodology

Bottom Line: Technology exists, physics supports detection at required ranges, and field operations have demonstrated success. Primary challenge is sedimentation (10+ cm/year in Mekong Delta), which actually makes magnetometry essential as optical/acoustic methods fail in buried scenarios.

1. TECHNICAL FOUNDATION

1.1 Detection Physics and Range Calculations

The fundamental principle governing magnetometer detection follows the magnetic dipole field equation.

Industry calibration standard, validated through thousands of field measurements: **1,000 kg of ferromagnetic material produces 1 nT anomaly at 30 meters distance**. This relationship scales predictably - doubling the distance reduces signal strength by a factor of eight, while doubling the mass only increases detection range by 26% (cube root relationship).

Target Component	Ferromagnetic Mass	Detection Range (0.25 nT)	Detection Range (1.0 nT)	Survey Line Spacing
Complete F-4 Phantom	3,500 kg	72 m	45 m	≤35 m
F-105 Thunderchief	3,800 kg	74 m	47 m	≤35 m
Twin J79 Engines	1,400 kg	54 m	34 m	≤27 m
Single Engine	700 kg	43 m	27 m	≤20 m
Landing Gear Assembly	900 kg	48 m	30 m	≤24 m
Structural Fragment	200 kg	29 m	18 m	≤14 m

1.2 Seawater Penetration Characteristics

Static magnetic fields penetrate seawater without attenuation. Unlike radar or ground-penetrating radar which suffer exponential signal loss in conductive seawater, magnetometer detection is affected only by geometric distance, not the intervening medium. Vietnam coastal waters (conductivity 4.5-5.5 S/m, temperature 25-30°C, salinity 32-35 PSU) present no barrier to magnetic field propagation.

Key Insight: A target at 50m depth below 50m of water produces identical magnetic signature as the same target at 50m distance in air. This makes shallow-water detection (0-100m) entirely feasible with surface-towed sensors, eliminating need for expensive deep-diving equipment.

2. FERROMAGNETIC COMPOSITION ANALYSIS

2.1 Material Science

Data reveals Vietnam-era aircraft contain substantial ferromagnetic materials concentrated in specific high-stress components. The aluminum airframe (70-80% of empty weight) contributes zero magnetic signature. However, critical systems require ferromagnetic materials for strength and heat resistance.

2.2 F-4 Phantom II Detailed Breakdown

System	Components & Materials	Ferromagnetic Mass	Detection Notes
J79-GE Engines (×2)	• 403 stainless steel compressor blades • Lapelloy rotor (steel alloy) • Steel casings and bearing housings • Note: Stages 1-7 use titanium (non-magnetic)	1,200-1,500 kg total	Strongest signature, concentrated mass
Landing Gear	• 300M high-strength steel struts • Steel wheels and brake assemblies • Hydraulic actuators	800-1,000 kg	Survives crashes intact
M61A1 Vulcan	• 6 rotating steel barrels • Steel housing and feed mechanism	112 kg	Compact, high-density target
External Stores	• Mk82 bombs (steel casings) • Rocket pods (steel tubes) • Fuel tank attachments	500-3,000 kg	Variable, mission-dependent
Structural	• Wing root fittings • Tailhook (Navy variants) • Control surface actuators	400-600 kg	Distributed throughout airframe
TOTAL	-	3,012-6,212 kg	21-43% of empty weight

2.3 Comparative Aircraft Analysis

- **F-105 Thunderchief:** Single J75 engine plus heavier structure = 3,360-4,160 kg ferromagnetic (28-34% of empty weight). The "Thud's" reputation for ruggedness translates to higher steel content.
- **A-4 Skyhawk:** Lightweight design = 1,460-1,910 kg ferromagnetic (31-40% of 4,750 kg). Higher percentage but lower absolute mass due to "Heinemann's Hot Rod" minimalist philosophy.
- **UH-1 Huey:** Transmission system dominates = 760-1,100 kg ferromagnetic (33-48% of 2,300 kg). Main rotor transmission contains 200-300 kg of solid steel gears - highest concentration in any component.
- **A-1 Skyraider:** Radial engine with steel crankcase = 1,800-2,200 kg ferromagnetic. Piston engines contain more ferromagnetic material than jets.

RECOMMENDED CONFIGURATION FOR OPERATIONS

Primary System: Geometrics G-882 Cesium Vapor Magnetometer

- Proven in 1,000+ marine surveys worldwide
- Detection range 48m for 1,000 kg target at professional threshold
- 20 Hz sampling enables 4-5 knot survey speeds
- GPS integration and MagLog software included
- Total system cost: \$45,000 (includes towfish, cable, software)

Budget Alternative: JW Fishers Proton 5 at \$20,843 provides adequate capability for major component detection with 30m range limitation.

3. SENSOR TECHNOLOGY EVALUATION

3.1 Available Systems Matrix

Technology/Model	Sensitivity	Sample Rate	Cost Range	Strengths	Limitations
Proton Precession JW Fishers Proton 5	1.0 nT	0.5 Hz	\$20,843	• Proven reliability • Simple operation • No dead zones	• Slow sampling • Limited to 3 knots
Cesium Vapor Geometrics G-882	0.004 nT	20 Hz	\$40-60K	• Industry standard • Excellent sensitivity • Fast sampling	• Dead zones at equator • Higher cost
Overhauser Marine Magnetics SeaSPY2	0.01 nT	4 Hz	\$35-55K	• No dead zones • Low power (3W) • Temperature stable	• Moderate sample rate • Less common
Fluxgate Bartington Grad601	0.1 nT	10 Hz	\$5-15K	• Very low cost • Compact size • Good for gradiometer	• Drift issues • Requires calibration

3.2 Performance Comparison at Vietnam Operational Depths

Testing data from similar tropical environments (South China Sea, Gulf of Thailand) demonstrates consistent performance across all magnetometer types at 0-100m depths. Key finding: water depth affects only tow cable management, not detection capability. Surface noise from waves becomes negligible beyond 10m cable length.

4. OPERATIONAL METHODOLOGY

4.1 Survey Design Optimization

Detection probability follows exponential decay based on line spacing versus detection radius:

$POD = 1 - \exp[-(r_{\text{max}}/s)^2]$ Where: POD = Probability of Detection r_{max} = Maximum detection radius s = Survey line spacing

This relationship drives survey planning. For 90% POD , line spacing must be $\leq 0.67 \times$ detection radius. Practical implementation:

- **Complete aircraft search:** 40m line spacing (covers 70m detection radius)
- **Engine/major components:** 25m line spacing (covers 45m detection radius)
- **Comprehensive debris field:** 15m line spacing (covers small fragments)

4.2 Environmental Factors Specific to Vietnam Waters

Sedimentation Challenge: Mekong Delta experiences 10+ cm/year accumulation rates. After 50-60 years, aircraft may be buried under 5-6 meters of sediment in worst cases, typically 0.5-3 meters. This burial makes magnetometer detection ESSENTIAL - side-scan sonar and optical methods cannot penetrate sediment cover.

Additional environmental considerations:

- **Visibility:** Gulf of Tonkin averages <1 foot, often 1-2 inches at depth (DPAA diving reports)
- **Monsoon patterns:** Northeast monsoon (Dec-Mar) creates 2-3m seas; plan operations Apr-Nov
- **Tidal currents:** 0.5-1.5 knots, requiring GPS-guided survey lines
- **Magnetic background:** Earth's field 40,000-43,000 nT at Vietnam latitudes (10-21°N)
- **Geology:** Basalt intrusions create 100-1,000 nT anomalies but have different signatures than aircraft

5. COST-BENEFIT ANALYSIS

5.1 System Acquisition Economics

Configuration	Capital Cost	Operating Cost/Day	Coverage Rate	Break-Even vs. Diving
Basic Proton System	\$25,000	\$500	2.0 km ² /day	8 days
Professional Cesium	\$50,000	\$750	2.4 km ² /day	12 days
AUV Platform	\$200,000	\$2,000	1.6 km ² /day	40 days
Diving Operations	N/A	\$8,000	0.01 km ² /day	Reference

Critical insight: Magnetometer surveys cost 10-20% of diving operations per unit area while covering 200× more seafloor. Even expensive AUV systems pay for themselves within two recovery missions.

5.2 Operational Efficiency Factors

- **Weather Independence:** Magnetometers operate in conditions that ground diving operations
- **Coverage Guarantee:** Systematic grid ensures no areas missed (unlike visual searches)
- **Data Permanence:** Digital records enable re-analysis with improved algorithms
- **Safety:** Zero diver exposure in contaminated or dangerous waters
- **Multi-tasking:** Can search while transiting between sites